Chapter

A Mathematical Model for Class Scheduling: A Case Study with the Federal University of Maranhão

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Abstract

Assigning university courses to classrooms is a recurring operational task that balances scarce space, laboratory requirements, and institutional inclusion policies. This chapter presents a mathematical programming approach to that task at the Federal University of Maranhão (UFMA), Brazil. After the weekly timetable is fixed, each meeting must receive a feasible room. We model the decision as an integer linear programme with two ordered goals: use as few rooms as possible, and, among such compact plans, reduce the placement of mobility-impaired student groups on upper floors. The chapter updates the educational timetabling and classroom-assignment literatures (including recent artificial intelligence [AI]-assisted scheduling), positions the contribution against prior CAP and fairness models, states a notated formulation with a justified lexicographic rule, and details a reproducible workflow with open artefacts. A case study at the Department of Informatics for the term 2023.1 shows that lexicographic optimization cuts room usage relative to both the historical hand-made plan and a constructive heuristic while improving the mobility criterion. Equally compact plans differ widely in accessibility, so an explicit second-stage optimization is necessary in practice. Additional experiments (weighted sums, disability and laboratory sensitivities, concurrent load, and five synthetic department profiles with 20 replicas each) confirm that the accessibility spread is structural. Open data and solver scripts accompany the chapter.

Keywords: class scheduling, classroom assignment, integer linear programming, university operations, accessibility, Federal University of Maranhão

1. Introduction

Every academic term, programme coordinators must decide where each class meeting will take place. Even when the times of meetings are already fixed (a common situation after the teaching timetable has been published), the remaining room-assignment step is nontrivial. Capacities must be respected, laboratories cannot host ordinary lectures (and vice versa), two meetings cannot share a room in

the same slot, and scarce rooms should not be wasted. In Brazilian public universities, these constraints coexist with legal and institutional duties toward students with disabilities, including prioritization of accessible spaces [1, 2].

Operations research has studied educational timetabling for decades [3–7]. Within that family, the classroom assignment problem (CAP) is the residual subproblem in which times (and often instructors) are given and only rooms remain to be chosen [8, 9]. Exact and heuristic methods are well established for compactness, preference, and distance objectives [9–11]. Recent surveys also document a growing interest in AI-assisted and hybrid solvers for university scheduling [12, 13]. What remains comparatively rare is an operational CAP model that treats physical mobility barriers as a first-class objective and that shows decision makers why a compact plan can still be inaccessible.

This chapter is a case study rather than a new named metaheuristic. We describe how CAP arises at the Federal University of Maranhão (UFMA), state a compact integer model aligned with departmental practice, and use lexicographic optimization to obtain plans that are lean in room usage and preferable for mobility-related disability enrolment. The managerial contribution is the accessibility gap among minimum-room optima: when many assignments attain the same $f_1^\star$, a compactness-only solver may return any of them. Measuring that range and optimizing the second stage is as important as writing the constraints.

The chapter is organized as follows. Section 2 reviews related work. Section 3 describes the UFMA setting and data. Section 4 presents the mathematical formulation. Section 5 explains the computational workflow. Section 6 reports the DEINF 2023.1 case study and additional sensitivity experiments. Section 7 extends the analysis with synthetic department profiles. Section 8 discusses practical recommendations, and Section 9 concludes.

2. Related literature

2.1 Educational timetabling and classroom assignment

Surveys of educational timetabling distinguish school, course, and examination settings and document a long tradition of exact, heuristic, and hybrid methods [3–7, 14–16]. Course timetabling typically assigns times and rooms jointly; examination timetabling emphasizes conflict graphs and seating. CAP arises when the timetable is already fixed and the remaining decision is a feasible room map [8, 17]. Classical complexity results place many timetable construction tasks among NP-hard problems [18], which motivate both sophisticated formulations and carefully engineered heuristics.

Integer programming has been applied successfully to university course and examination timetabling [10, 11, 19–21]. For classroom assignment specifically, Phillips et al. [9] develop IP methods for large-scale practical instances and emphasize computational tractability once times are given. Preference-based and distance-based objectives appear frequently in CAP and related studies [9, 22–25]. Metaheuristics remain competitive for integrated timetabling [14, 26–28], but exact methods are attractive for post-timetable CAP because the fixed timetable yields moderate binary models that modern mixed integer programming (MIP) solvers often close in seconds at departmental scale.

Recent benchmark-oriented overviews emphasize reproducibility and standard formulations [7, 29], while strategic and multi-criteria university planning continue to attract attention [30, 31]. Gu et al. [12] review integer-programming and machine-learning approaches for university timetabling and highlight hybrid IP–ML pipelines as a near-term research direction. Complementary practice-facing work uses knowledge-representation tools such as answer-set programming with web interfaces for curriculum-based course timetabling [13], and recent faculty-scale IPs incorporate room stability and lecturer preference [32]. These streams improve automation and preference handling; they do not, as a rule, optimize an explicit mobility-floor penalty after the timetable is fixed.

2.2 Fairness, equity, and accessibility

Equity and fairness criteria have received growing attention in operations research [33–37]. In educational timetabling, fairness is often framed as balanced distribution of soft-constraint discomfort among students or teachers [7, 38], rather than as physical building accessibility. In facility and service systems, accessibility is modelled through distance, capacity, or demographic coverage [39]. Campus architecture research likewise stresses universal design of circulation and teaching spaces [1, 40], but typically as a building-assessment agenda rather than as a term-by-term assignment model. In educational buildings, floor level remains a practical proxy for mobility barriers when elevators are limited or unreliable, a situation common in many public universities.

Despite this, CAP models rarely treat disability-related accessibility as a primary objective. Closest in spirit are studies that soft-constrain proximity to accessible buildings [41], without optimizing an explicit mobility penalty or analysing alternate optima on a compactness face. Multi-objective optimization offers hierarchical and scalarization tools for ordered or conflicting criteria [35, 42–44]. Lexicographic optimization is appropriate when institutional priorities are ordered: room scarcity is binding, but accessibility must be optimized among compact plans.

2.3 Positioning of this chapter

Table 1 summarizes representative strands and the gap addressed here. Mature CAP and fairness literatures exist, yet few operational studies (1) optimize an explicit floor-based mobility penalty jointly with compactness, (2) quantify the accessibility range among minimum-room optima beyond the two extremes, and (3) provide a diagnostic workflow that estimates the risk of landing on an inaccessible compact plan. This chapter fills that gap in a UFMA case-study format, with open artefacts and an emphasis on managerial interpretation.

3. Institutional setting and data

UFMA is a multi-campus federal university. Room allocation is repeated each academic period and, in several units, is still performed largely by hand, using spreadsheets and informal negotiation among coordinators. Resolution No. 2,858-CONSEPE/2023 requires prioritization of accessible physical spaces for students with disabilities, with particular attention to users of orthoses or prostheses

Strand	Typical focus	Accessibility	Relation to this chapter
Classical CAP/IP [9, 10]	Compactness, preferences, tractability	Rarely primary	We keep a compact IP but add a mobility stage
Integrated timetabling heuristics [26, 28]	Joint time–room search	Soft preferences, not floors	We assume times are fixed (post-timetable CAP)
Fairness in timetabling [35, 38]	Balanced soft-constraint burden	Equity among agents, not floors	Complementary; different criteria
AI/hybrid scheduling [12, 13]	Automation, ML guidance, ASP tools	Not mobility-floor CAP	Useful for larger CTT; not for our residual CAP
Campus universal design [1, 40]	Building and circulation quality	Physical inclusion by design	Complements the assignment; does not assign rooms
This chapter	Lexicographic CAP at UFMA	Explicit f_2 on compact face	Gap diagnostics and reproducible workflow

Table 1.
Related approaches and the gap addressed in this chapter.

[2]. That regulation motivates treating floor level as a proxy for vertical mobility barriers when elevators are limited or unreliable, a frequent situation in older academic buildings on the São Luís campus.

Our case focuses on the Department of Informatics (DEINF), located in the Centre for Exact Sciences and Technology (CCET). Computer Science runs as an afternoon/evening programme (roughly 14:00–22:00, Monday to Friday). Meetings may use rooms in DEINF itself or in the neighbouring Paulo Freire pedagogical building. The university IT unit (STI) provides meeting-level flags for visual and physical disability enrolment without identifying individual students. Because upper floors primarily hinder mobility, the base study marks a meeting as mobility-relevant ($PWD_d = 1$) only when at least one enrolled student has a physical/mobility disability. A sensitivity check later replaces that rule by the broader 'any disability' OR of the STI flags.

For term 2023.1, the processed instance contains 64 course-meeting entries, 25 coded time slots, and 31 rooms (29 physical rooms plus two synthetic higher-floor laboratories added to mirror the departmental practice of borrowing research labs when concurrency exceeds the four official computer laboratories). Six meetings carry a mobility flag under the base definition. The historical manual assignment occupied 23 rooms. Disability attributes are protected under Brazil's LGPD [45]. Released files retain only anonymized meeting codes, aggregate enrolment, binary mobility flags, and room attributes; no student identifiers or diagnoses are published.

4. Mathematical model

Table 2 collects the notation used below. Let D be the set of meetings, T the discrete time slots, $T_d \subseteq T$ the slots of meeting d, and S the room pool. Eligibility $C_{ds} \in \{0, 1\}$ encodes capacity and laboratory compatibility. Parameter $Floor_s \in \mathbb{Z}_{\geq 0}$ is the floor of room s (ground floor equals zero). Binary variables X_{dts} indicate that meeting d uses room s in slot t, and Y_s indicates that room s is used at least once. Variables X_{dts} exist only for feasible triples (d, t, s).

Symbol	Meaning
D, T, S	Meetings, time slots, rooms
$T_d \subseteq T$	Slots required by meeting d
C_{ds}	Eligibility of room s for meeting d (capacity and laboratory type)
$Floor_s$	Floor index of room s (0 = ground)
PWD_d	1 if meeting d has mobility-relevant enrolment
X_{dts}	1 if meeting d uses room s in slot t
Y_s	1 if room s is used at least once
$f_1(Y)$	Number of rooms used (compactness)
$f_2(X)$	Floor penalty accumulated by mobility-relevant meetings
$f_1^\star$	Minimum attainable room count
$f_2, \bar{f}_2, \Delta$	Best/worst mobility penalty on the compact face, and their gap

Table 2.
Notation used in the classroom-assignment model.

$$\sum_{t \in T_d} \sum_{s \in S} X_{dts} = |T_d| \qquad d \in D, \tag{1}$$

$$\sum_{d \in D} X_{dts} \le 1 \qquad t \in T, \quad s \in S, \tag{2}$$

$$\sum_{s \in S} X_{dts} \le 1 \qquad d \in D, \quad t \in T_d, \tag{3}$$

$$X_{dts} \le Y_s \qquad d \in D, \quad t \in T_d, \quad s \in S, \tag{4}$$

Constraint (1) covers every meeting slot, (2) forbids double booking, (3) assigns at most one room per meeting–slot, and (4) activates room-usage indicators. The formulation follows the classical post-timetable CAP structure used in integer-programming classroom assignment [9, 10], with eligibility restricted a priori by C_{ds}.

Two objective functions capture the institutional priority order:

$$f_1(Y) = \sum_{s \in S} Y_s, \tag{5}$$

$$f_2(X) = \sum_{d \in D} \sum_{t \in T_d} \sum_{s \in S} X_{dts}\, Floor_s\, PWD_d. \tag{6}$$

Here f_1 measures compactness and f_2 accumulates a linear floor penalty only for mobility-relevant meetings. The decision problem solved in the case study is the lexicographic programme

$$\text{lexmin}\,(f_1, f_2) \quad \text{subject to}\,(1) \quad -(4). \tag{7}$$

Equivalently, one may first minimize f_1, then minimize f_2 subject to $f_1 \le f_1^\star$ [42, 43]. We prefer lexicographic optimization over a weighted sum or an ε-constraint

trade-off for three reasons that match UFMA practice. First, coordinators state an ordered priority (free rooms before refining inclusion), not a calibrated trade-off weight. Second, on the DEINF instance every tested weighted sum with $\alpha > 0$ recovers the same compact accessible point (Section 6), so scalarization adds little managerial information once accessibility enters the model. Third, lexicographic optimization guarantees that accessibility is optimized on the entire minimum-room face, which is exactly where the accessibility gap Δ appears. Weighted sums remain useful for illustration and sensitivity plots; they are not required to implement the recommended operational rule. If a calibrated accessibility score $A_s \in [0,1]$ becomes available, $Floor_s$ in (6) can be replaced by $1 - A_s$ without changing the constraint system.

Let $\mathcal{X}_1^\star$ be the set of feasible assignments with $f_1 = f_1^\star$, and write

$$\underline{f}_2 = \min_{X \in \mathcal{X}_1^\star} f_2(X), \qquad \overline{f}_2 = \max_{X \in \mathcal{X}_1^\star} f_2(X), \qquad \Delta = \overline{f}_2 - \underline{f}_2.$$

The gap Δ is the accessibility spread among equally compact plans. Whenever $\Delta > 0$, compactness alone does not determine an inclusive schedule. The relative gap $\rho = \Delta / \overline{f}_2$ (with $\rho := 0$ if $\overline{f}_2 = 0$) normalizes that spread for communication with non-specialists.

5. Computational workflow

The case study uses a transparent three-stage workflow that departmental staff can reproduce with open-source tools (Algorithm 1). The stages follow standard lexicographic multi-objective practice [42, 43] and the CAP modelling tradition of [9].

Compactness. Solve $\min f_1$ subject to (1)–(4) and record $f_1^\star$.

Accessibility extremes on the compact face. Minimize and maximize f_2 under the additional inequality $f_1 \le f_1^\star$, obtaining $\underline{f}_2, \overline{f}_2$, and Δ.

Sampling of alternate compact plans. Draw N members of $\mathcal{X}_1^\star$ by minimizing a random linear cost

$$\sum_{s \in S} w_s Y_s + \varepsilon \sum_{(d,t,s)} u_{dts} X_{dts}, \tag{8}$$

with $w_s, u_{dts} : U(0,1)$ and a small $\varepsilon > 0$ (we use 10^{-4}). From the sample one reports the mean penalty $\overline{f}_2$, the hit rate of the accessible extreme

$$h^\star = \frac{1}{N} \left| \left\{ k : f_2^{(k)} = \underline{f}_2 \right\} \right|,$$

and a normalized mean excess $\eta = (\overline{f}2 - \underline{f}_2)/\Delta$ (with the convention $\eta = 0$ if $\Delta = 0$).

If $\Delta > 0$ and random compact optima rarely hit $\underline{f}_2$, the recommended operational rule is precisely (7). As baselines we keep the historical manual plan and a greedy rule that, slot by slot, chooses an eligible free room of smallest capacity, breaking ties toward lower floors for mobility-relevant meetings. We deliberately compare against a constructive heuristic and against institutional practice rather than against a large suite of metaheuristics: at departmental scale the MIP closes in seconds, so the scientific question is whether accessibility enters the model, not whether a genetic algorithm can approximate the same optimum.

Task	Rooms/penalty	CPU(s)
Lexicographic ILP (reported plan)	$f_1 = 13, f_2 = 8$	≈ 3.5
Face extremes (min/max f_2 at $f_1^\star$)	$\Delta = 16$	≈ 3.6
Greedy-constructive baseline	$f_1 = 20, f_2 = 11$	< 0.1

Table 3.
Illustrative computational effort on DEINF 2023.1 (highs via PuLP).

All MIPs are implemented in Python with PuLP and solved by HiGHS [46] (CBC as fallback) under a 60-second time limit. Experiments were run on a workstation with Ubuntu Linux, an Intel Core i5 processor, and 32 GB of RAM. After domain reduction the DEINF model has on the order of 1.5×10^3 binary variables and 2×10^3 constraints. **Table 3** reports two separate MIP tasks on DEINF: the lexicographic plan that yields $(f_1, f_2) = (13, 8)$ takes about 3.5 seconds, and computing the compact-face extremes $\underline{f}_2$ and $\overline{f}_2$ (hence Δ) takes about 3.6 seconds. Those times are of the same order and are not additive stages of a single stopwatch reading; either diagnostic finishes in a few seconds. Synthetic 100-meeting pools remain under roughly 12 seconds on average (Section 7).

5.1 Reproducibility

Anonymized databases (meetings, rooms, slots, eligibility), solver scripts, and result tables are released with the chapter (see Data availability). Peers can regenerate **Tables 4–6** and **Figures 1–7** by installing the Python dependencies listed in the repository and running the documented entry-point scripts under a 60-second MIP time limit. Disability attributes remain binary meeting-level flags only, consistent with LGPD constraints [45].

Algorithm 1: Practical workflow for accessibility-aware classroom assignment

Require: Instance (D, T, S), sample size N
Ensure: compact, accessible assignment and diagnostic summary.
$f_1^\star \leftarrow \min\{ f_1 : (1) \quad -(4)\}$
$\underline{f}_2 \leftarrow \min\{ f_2 : (1) \quad -(4), f_1 \le f_1^\star\}$
$\overline{f}_2 \leftarrow \max\{ f_2 : (1) \quad -(4), f_1 \le f_1^\star\}$
$\Delta \leftarrow \overline{f}_2 - \underline{f}_2$
for $k = 1, \ldots, N$ **do**
    Sample random weights and solve (8) on the face $f_1 \le f_1^\star$
    Record $f_2^{(k)}$
end for
Solve lexmin(f_1, f_2) (or minimize f_2 subject to $f_1 \le f_1^\star$)
return assignment, Δ, and sample statistics

Policy	Rooms used	Mobility penalty	Reduction versus manual
Manual	23	N/A	N/A
Greedy	20	11	13%
Lexicographic ILP	13	8	43.5%

Table 4.
DEINF 2023.1: manual practice, greedy construction, and lexicographic ILP.

Scenario	$f_1^\star$	Lex f_2	$\underline{f}2$	$\bar{f}_2$	Status
Mobility PWD, base labs	13	8	8	24	Optimal
Any disability (STI OR)	13	8	8	30	Optimal
Synthetic labs on the ground floor	13	8	8	24	Optimal
No synthetic labs	–	–	–	–	Infeasible

Table 5.
Sensitivity on DEINF-2023.1.

Profile	$f_1^\star$	$\underline{f}2$	Δ	CPU(s)
DEMAT	10.8 ± 1.5	1.1 ± 2.8	33.8 ± 6.8	2.7 ± 2.3
DEEE	13.5 ± 1.3	0.2 ± 0.6	39.3 ± 8.5	3.6 ± 2.7
DEFI	12.8 ± 1.7	0.8 ± 1.7	27.5 ± 6.7	2.5 ± 1.7
CCET-large	17.6 ± 2.2	3.0 ± 3.8	79.2 ± 11.8	11.6 ± 6.5
HighPWD	13.1 ± 1.7	3.4 ± 3.3	85.7 ± 12.7	3.2 ± 2.1

Table 6.
Synthetic profiles (20 seeds). values are mean ± standard deviation.

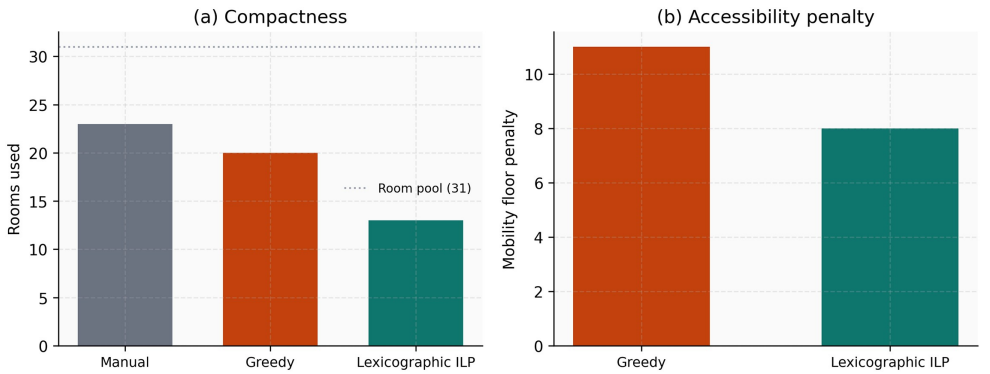


Figure 1.
Compactness and mobility penalty under three policies for DEINF 2023.1.

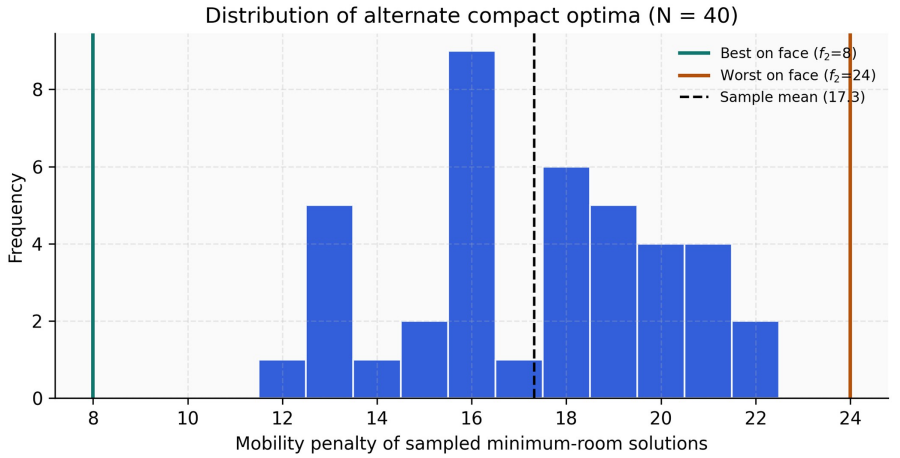


Figure 2.
Distribution of mobility penalties among 40 randomly sampled 13-room optima.

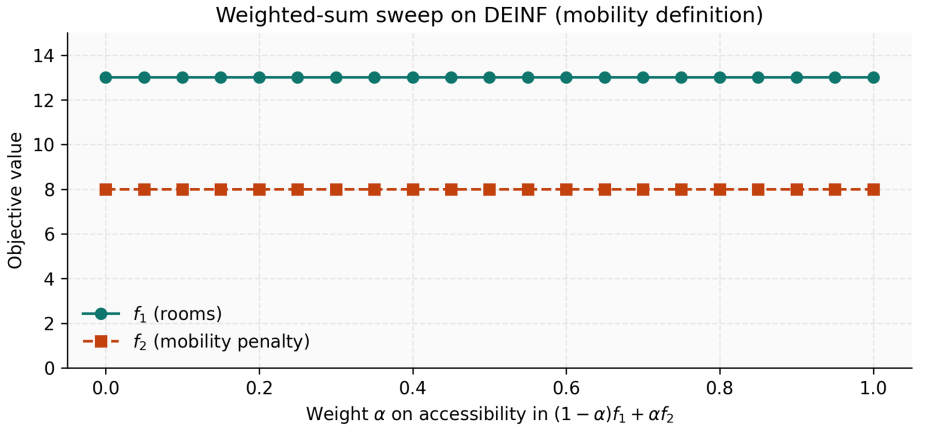


Figure 3.
Weighted-sum sweep on DEINF under the mobility definition of PWD_d.

6. Case study: DEINF, term 2023.1

6.1 How much can optimization help?

We first compare three operational policies on the same DEINF 2023.1 instance: the historical manual assignment, a constructive greedy heuristic, and the lexicographic ILP. **Table 4** summarizes the headline metrics. The manual plan occupies 23 of the 31 available rooms and does not record a mobility penalty in the historical spreadsheet. Greedy already improves compactness to 20 rooms (a 13% reduction) with a mobility penalty of 11, which shows that simple constructive rules capture part of the structure. Lexicographic optimization, however, reaches 13 rooms and a penalty of 8. Relative to the manual baseline, this is a 43.5% reduction in

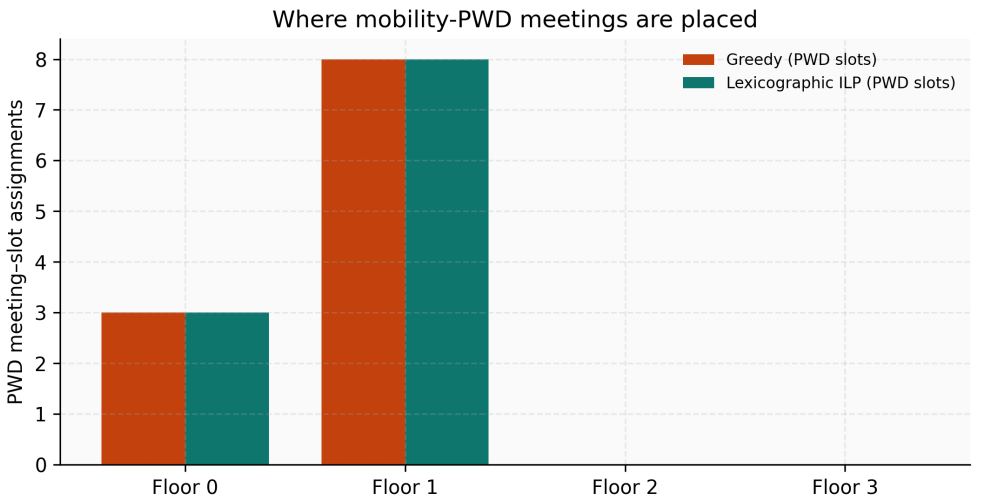


Figure 4.
Floor placement of mobility-PWD meeting slots ($PWD_d = 1$): greedy versus lexicographic ILP.

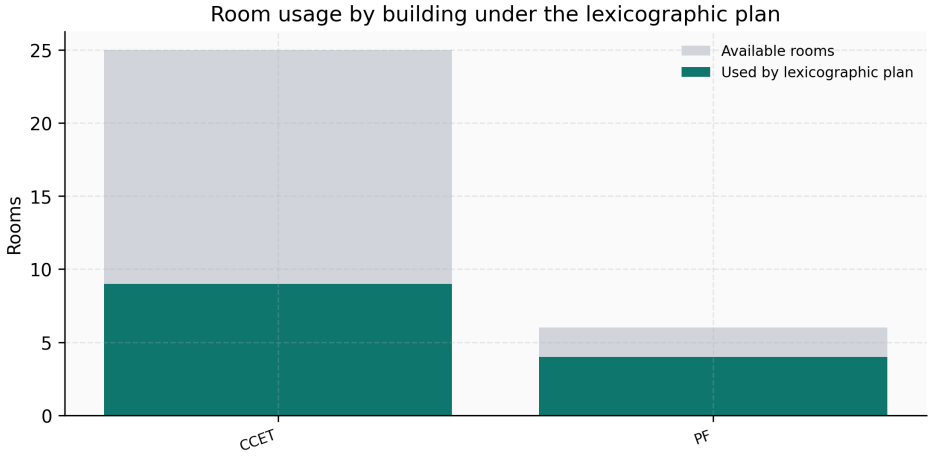


Figure 5.
Rooms available and rooms used by building under the lexicographic assignment.

room usage; relative to greedy, it is both more compact and more accessible. Solve times remain at a few seconds on an Ubuntu i5 workstation with 32 GB of RAM, so the gain is not purchased at the cost of an overnight run.

Reading **Table 4** carefully, two managerial conclusions stand out. First, hand-made practice leaves a large compactness gap: 10 rooms are 'saved' only when the ILP is used, which frees teaching space for other programmes or maintenance. Second, accessibility does not deteriorate when compactness improves. On the contrary, the lexicographic plan dominates greedy in both columns. That pattern is visualized in **Figure 1**. **Figure 1a** places the three policies against the full pool of 31 rooms; the dashed reference line makes the residual unused capacity explicit.

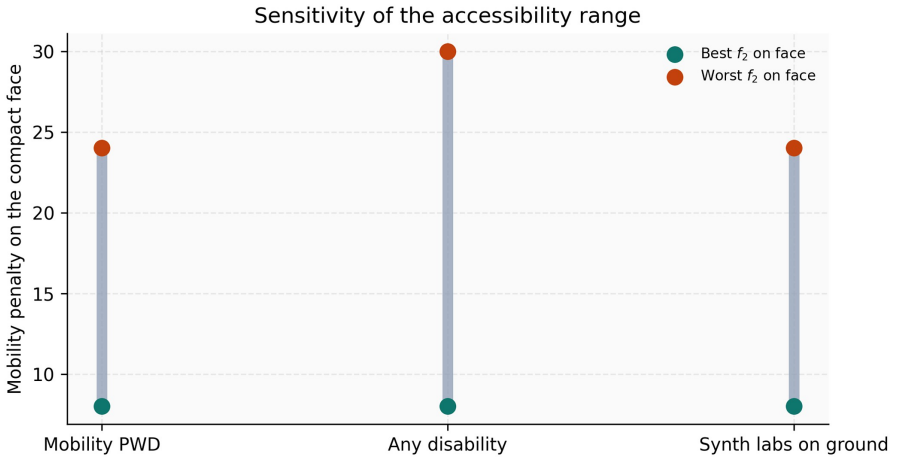


Figure 6.
Best and worst mobility penalties on the compact face under sensitivity scenarios.

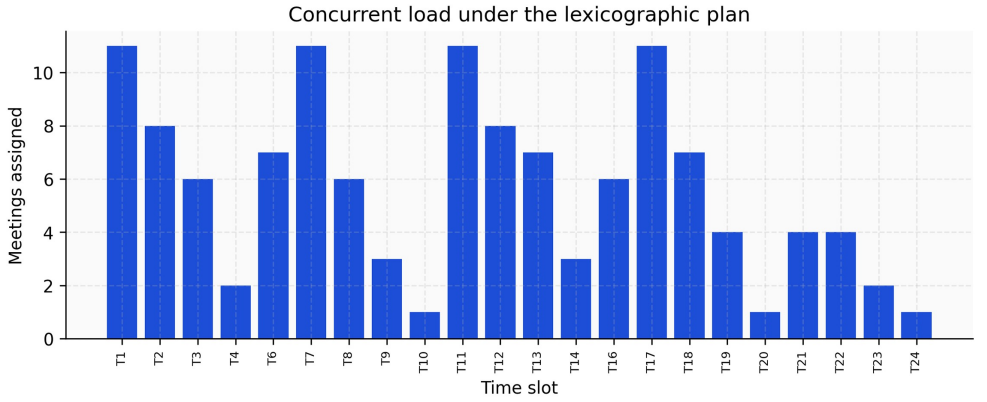


Figure 7.
Concurrent meeting load by time slot under the lexicographic assignment.

Figure 1b shows that the mobility penalty falls from 11 under greedy to 8 under the ILP. Taken together, the figure argues against the informal belief that inclusion must inflate the room footprint: at DEINF, better modelling improves both axes at once.

6.2 Why the second objective matters

Minimizing f_1 alone already yields $f_1^\star = 13$. That optimum, however, is not unique in the accessibility dimension: on the compact face one finds $f_2 \in [8, 24]$, so $\Delta = 16$ and $\rho = 2/3$. In other words, two plans that look identical to a compactness-only dashboard can differ by 16 penalty units for mobility-relevant meetings.

Sampling 40 alternate compact optima by (8) produces the distribution in **Figure 2**. The histogram is centred well above the accessible extreme: the sample mean is 17.3 (standard deviation 2.8; median 18; sample range 12–22). The vertical

line at $f_2 = 8$ marks the best compact plan, and the line at $f_2 = 24$ marks the worst; none of the 40 draws lands on the best value ($h^\star = 0$). About 78% of the sample lies on or above the mid-face threshold, and the normalized mean excess is $\eta \approx 0.58$. The figure therefore does more than display variability: it quantifies operational risk. If a departmental system returns 'any' 13-room solution, the expected mobility outcome is closer to the middle of the face than to the inclusive extreme, even though an equally compact inclusive plan exists.

6.3 Weighted sums and the absence of a first-best trade-off

A natural question is whether accessibility must be purchased by using more rooms. **Figure 3** answers that question for DEINF under a classical weighted sum $(1-\alpha)f_1 + \alpha f_2$ with $\alpha \in \{0, 0.05, \ldots, 1\}$. Both curves are flat at $(f_1, f_2) = (13, 8)$ for every tested weight, including the lexicographic refinements at the endpoints. The plot should be read as a negative result of practical importance: once accessibility receives any positive weight, the solver recovers the same compact accessible point. There is no efficiency–equity frontier to navigate at the first-best room count. The conflict appears only when f_2 is omitted and any member of $\boldsymbol{X}_1^\star$ may be selected [33–35]. For coordinators, the implication is that the hard step is not choosing α; it is ensuring that accessibility enters the model at all.

6.4 Where meetings and mobility-relevant groups go

Aggregate penalties hide the spatial mechanism. **Figure 4** decomposes slots of meetings with $PWD_d = 1$ by floor under greedy and under the lexicographic ILP. Greedy already tilts toward the ground floor, which confirms that a local preference for lower floors is easy to encode heuristically. Nevertheless, residual mobility-PWD slots remain on upper floors, and those residuals accumulate into the penalty of 11. The lexicographic bars shift mass further toward lower levels while still respecting laboratory and capacity constraints. The figure clarifies that inclusion is achieved by selective prioritization of meetings with $PWD_d = 1$, not by forcing every class onto the ground floor.

Figure 5 examines the same lexicographic plan from the building side. For each building in the pool, the light bar is the number of available rooms and the dark bar is the number actually used. The optimized assignment draws rooms from more than one building, which matches the operational reality of borrowing adjacent teaching spaces (including Paulo Freire) when DEINF capacity alone is insufficient. The unused remainder within buildings is the compactness dividend already seen in **Table 4**: rooms are released without concentrating all traffic into a single block.

6.5 Sensitivity to disability definition and laboratories

Robustness checks ask whether the qualitative message depends on modelling details. **Table 5** reports four scenarios. The mobility base case recovers $(f_1, f_2) = (13, 8)$ with face range [8, 24]. Expanding the disability flag to 'any STI disability' (seven meetings instead of six) keeps the lexicographic point at $(13, 8)$ but widens the worst compact plan to $f_2 = 30$. That widening is informative: if visual-only enrolment is incorrectly treated as a floor penalty, the inaccessible extreme of the face becomes even worse, while the recommended plan remains unchanged.

Placing the two synthetic laboratories on the ground floor leaves both the lexicographic optimum and $\Delta = 16$ unchanged, because physical ground-floor labs are already preferred when free; the synthetic labs mainly restore feasibility. Removing them makes the MIP infeasible, which isolates laboratory concurrency (not the optimizer) as the binding resource.

Overall, **Table 5** supports three claims used later in the recommendations: (1) the accessible compact plan is stable across reasonable disability definitions; (2) the accessibility gap on the face remains strictly positive whenever the model is feasible; and (3) laboratory stock, not algorithmic weakness, explains hard infeasibility. **Figure 6** restates the same message graphically by plotting best and worst face penalties under the feasible scenarios: the lower marker stays at 8, while the upper marker rises when the disability flag is broadened.

6.6 Concurrent load and classroom utilization

Coordinators often ask whether a compact plan creates hidden congestion in peak slots. **Figure 7** plots the number of meetings assigned in each time slot under the lexicographic solution. The load varies across the week, with a few denser afternoon/evening blocks that still remain within the available room pool once laboratories and capacities are respected. The figure is therefore a utilization check rather than an additional objective: it shows that reducing the room footprint from 23 to 13 does not force an infeasible spike in any single slot in DEINF 2023.1. Together with **Figure 5**, it gives a more concrete picture of classroom utilization than aggregate room counts alone.

7. Beyond a single timetable

Because comparable archives for neighbouring departments were unavailable under the same schema, we generated five synthetic profiles calibrated on DEINF statistics and replicated each profile over 20 random seeds with a capacity-repair step that yields MIP-feasible instances. Profiles deliberately vary laboratory intensity and the rate of mobility-PWD meetings ($PWD_d = 1$) rather than cloning the empirical DEINF distribution; they are stress tests, not substitutes for unpublished departmental archives.

Table 6 reports mean ± standard deviation for $f_1^\star$, $\underline{f}_2$, Δ, and CPU. Every feasible replica has $\Delta > 0$, so the DEINF phenomenon is not an artefact of one timetable seed. Mean $\underline{f}_2$ stays comparatively low (often near zero), which means an accessible compact plan usually exists. What varies is how bad the worst compact plan can be: the lightest profile, DEFI, shows mean gaps near 27; DEMAT sits in a mid band near 34; laboratory-heavy DEEE, the large CCET pool, and especially HighPWD reach mean gaps of roughly 39 to 86. CPU remains a few seconds on departmental sizes and about 12 seconds on the 100-meeting pool, still compatible with pre-term planning.

Figure 8 plots the mean accessibility gap with error bars equal to one standard deviation. The ordering of the bars mirrors the stress built into each profile: HighPWD and CCET-large dominate; DEEE and DEMAT occupy intermediate bands (laboratory concurrency for DEEE, a milder mid-gap for DEMAT); and DEFI is the mildest. The error bars are non-trivial but never cross zero, which strengthens the claim that a positive gap is structural under the generator. For a coordinator, the

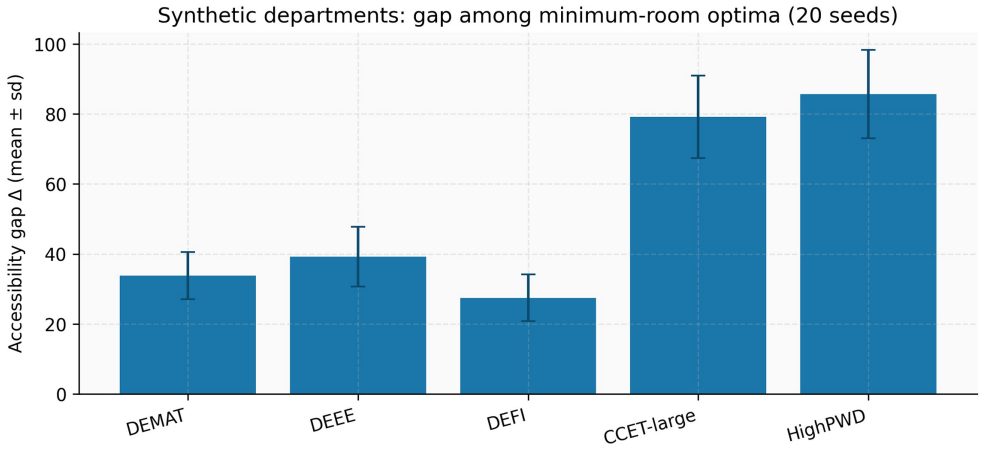


Figure 8.
Mean accessibility gap Δ across synthetic department profiles.

practical reading is that departments with denser laboratory demand or higher mobility-PWD load should expect a wider compact face and therefore a higher payoff from the second lexicographic stage. **Figure 9** relates replica-level gaps to solve times: larger gaps are not systematically associated with prohibitive runtimes at the tested sizes, which supports routine use of the face diagnostic before publishing a room map.

Figure 10 reorganizes the same evidence in the plane of mean room usage versus mean gap, with the real DEINF point marked as a diamond. The scatter shows that higher compactness pressure (larger $f_1^\star$) often co-moves with larger accessibility diversity, but the relationship is not a simple line: HighPWD has a moderate room count yet the largest gap, because many mobility-relevant meetings amplify f_2 differences among compact plans. DEINF sits toward the lower-left relative to the

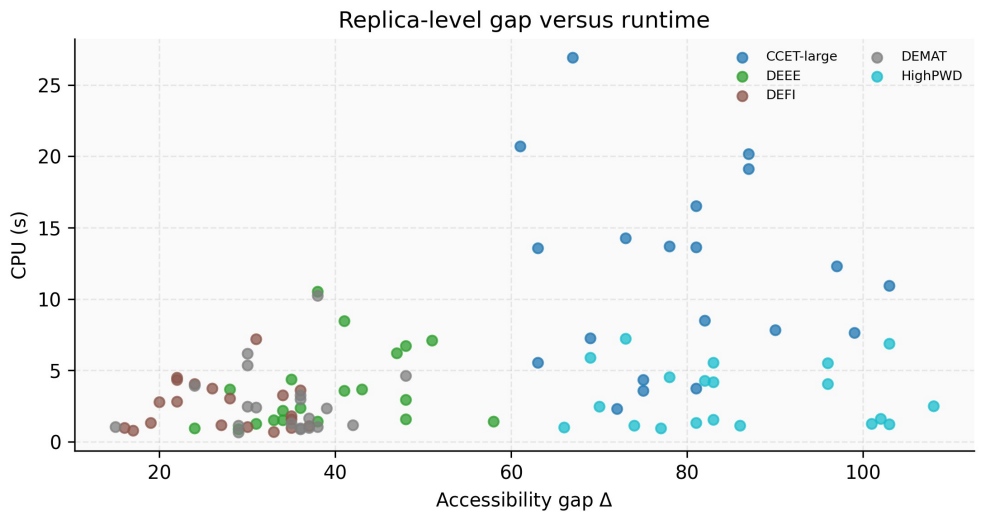


Figure 9.
Replica-level accessibility gap versus CPU time across synthetic profiles.

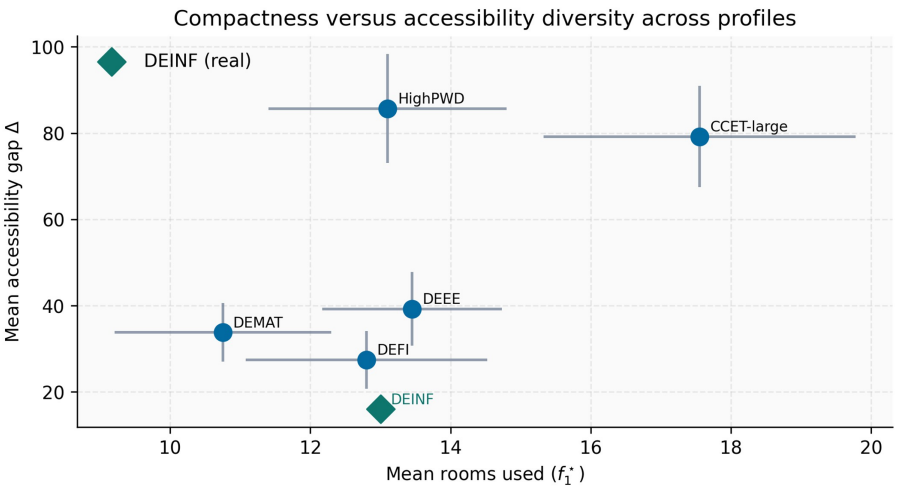


Figure 10.
Compactness versus accessibility diversity for synthetic profiles and the real DEINF instance.

synthetic stress tests, which suggests that the real case is not an extreme outlier; if anything, the synthetic profiles explore harsher regimes. The figure therefore supports external validity in the weak sense that the accessibility-gap mechanism survives under varied scale and composition.

8. Practical recommendations for coordinators

The UFMA case suggests a short checklist that other departments can adopt without specialized OR staff.

(1) Keep the timetable fixed and export meetings, slots, capacities, laboratory flags, floors, and anonymized mobility indicators.

(2) Compute $f_1^\star$ and the face extremes $\underline{f}_2, \overline{f}_2$. If $\Delta = 0$, compactness already determines accessibility; otherwise continue.

(3) Inspect a small random sample of compact optima. If $h^\star \approx 0$ and η is large, do not publish a compactness-only plan.

(4) Publish the lexicographic plan, not an arbitrary minimum-room solution.

(5) Treat laboratories as the scarce resource: infeasibility often signals missing lab capacity rather than a modelling error.

(6) Document the mobility definition used (physical-only versus any disability flag) so that inclusion reports remain auditable under institutional policy.

On DEINF, inclusion did not require additional rooms: the accessible compact plan and the least accessible compact plan both used 13 rooms. The cost of inclusion

was computational (a second MIP stage of a few seconds) rather than physical expansion of the room stock. That distinction matters for budget discussions: accessibility compliance can often be improved by better assignment, not only by construction projects.

8.1 Implementation challenges

In practice, the hardest steps are organizational rather than mathematical. STI extracts must be aligned with the room inventory (including borrowed spaces such as Paulo Freire), and disability flags must remain anonymized at meeting level under the LGPD [45]. Coordinators also need a clear escalation path when the MIP reports infeasibility: the DEINF experiments show that missing laboratory capacity is a frequent root cause. Finally, publishing only one compact plan without the face range can create a false sense of compliance, because $\Delta > 0$ means that another equally compact plan may place mobility-relevant groups on higher floors.

9. Conclusions

This chapter presented a case study of classroom assignment at UFMA through a lexicographic integer linear programme that first minimizes room usage and then a floor-based mobility penalty. After reviewing educational timetabling, CAP, fairness, campus universal design, and recent AI-assisted scheduling work, we argued that physical mobility accessibility remains under-represented as a primary CAP objective. On the DEINF 2023.1 instance, the optimized plan uses 13 rooms with mobility penalty of 8, improving on both the historical manual assignment and a greedy heuristic. Equally compact plans span penalties from 8 to 24, and random sampling of that face concentrates away from the accessible extreme. Weighted-sum and sensitivity experiments clarify that the issue is alternate optima on a compact face, not a classical trade-off at the first-best room count. Synthetic departmental profiles confirm that a positive accessibility gap is the rule rather than the exception, while CPU times remain compatible with pre-term planning.

9.1 Managerial implications

For programme coordinators and campus operations, three messages matter. First, inclusion need not inflate the room footprint when accessibility is optimized on the compact face. Second, dashboards that report only the number of rooms used are incomplete whenever $\Delta > 0$: they should also report $\underline{f}_2, \overline{f}_2$, and preferably a small sample of alternate compact plans. Third, laboratory capacity is often the true scarce resource; investment and borrowing decisions should be informed by infeasibility diagnostics, not only by average occupancy.

9.2 Limitations

The floor proxy does not encode elevator reliability, ramp continuity, or walking paths between buildings. The model omits room-stability preferences across consecutive slots and multi-campus travel times. Disability flags are meeting-level aggregates, which are appropriate under privacy rules but coarser than

individualized accommodation plans. The synthetic profiles stress-test the mechanism rather than replace unpublished departmental archives, and we did not run a large horse-race of metaheuristics because departmental MIPs close quickly.

9.3 Future research

Natural extensions include multi-campus benchmarks with richer accessibility scores A_s, robust or stochastic models under enrolment uncertainty, ε-constraint analyses for additional soft preferences (room stability, lecturer buildings), and hybrid AI-assisted pipelines that learn eligibility or demand patterns while retaining an exact lexicographic CAP core [12]. Lightweight interfaces that run Algorithm 1 before each term's room map is published would help transfer the method beyond DEINF.

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Conflict of Interest

The authors declare no conflict of interest.

Data availability

Anonymized instance files, result tables, and solver scripts are available at https://github.com/glaubosclimaco/accessibility-aware-classroom-assignment (directory code/).

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